S07 reachable near-drop preparation analysis

Documentary and analytical review artifact

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1 Question and claim boundary

The accepted task is a 2,400 m three-lane approach followed by a three-to-two lane reduction. Common desired-speed advice applies on $[800, 2000]$ m; the hidden-lane contrast is required in the final 400 m, $R = [2000, 2400]$ m, immediately before the narrowing. The preparation starts from an empty road and the strong lane-blind channel retains complete spatial count, density, mean-speed, cumulative-count, source- waiting, and advice histories through decision time.

This analysis asks when a nontrivial contrast can reach R, how a small outcome-blind bank can avoid obvious insertion confounding, and why exact aggregate-history equality still cannot be certified without a native reference run. It does not supply such a run, approve revised settings, or establish that lane information changes useful advice.

2 Arrival bounds

Let the default base insertion put a 5 m vehicle's front at $x_0 = 5$ m. Define

$$x_z = 800\,\text{m}, \qquad x_R = 2000\,\text{m}, \qquad v_{100} = \frac{100}{3.6} = 27.7778\,\text{m/s}, \qquad v_{95} = \frac{95}{3.6} = 26.3889\,\text{m/s},$$

where 95 km/h is the nominal free speed induced by baseline advice 90 km/h and response gain 0.5. The declared acceleration is $a = 2.6\,\text{m/s}^2$.

Two strict kinematic screens ignore all safety, interaction, insertion, and advice delays. Instantaneous travel at v_{100} gives

$$\tau_{\text{inst}} = \frac{x_R - x_0}{v_{100}} = 71.8200\,\text{s}.$$

Starting from rest, accelerating to v_{100}, and then maintaining it gives

$$\tau_{\text{acc}} = \frac{v_{100}}{a} + \frac{x_R - x_0 - v_{100}^2/(2a)}{v_{100}} = 77.1619\,\text{s}.$$

These reproduce S06's robust exclusion of the 40 s and 60 s decisions.

For planning rather than proof, two nominal free-flow travel times respect the 95 km/h advice-zone speed. Insertion at desired freeway speed gives

$$\tau_{\text{des}} = \frac{x_z - x_0}{v_{100}} + \frac{x_R - x_z}{v_{95}} = 74.0937\,\text{s}.$$

Insertion from rest, acceleration to v_{100} before x_z, and nominal v_{95} travel through the advice zone gives

$$\tau_0 = \frac{v_{100}}{a} + \frac{x_z - x_0 - v_{100}^2/(2a)}{v_{100}} + \frac{x_R - x_z}{v_{95}} = 79.4356\,\text{s}.$$

The latter is not a guaranteed upper bound: following, insertion delay, lane changing, and downstream interaction can make travel later.

After x_R, advice no longer lowers desired speed. Nominal acceleration from v_{95} to v_{100} gives a region residence estimate

$$\delta_R = \frac{v_{100} - v_{95}}{a} + \frac{400 - \frac{1}{2}(v_{95} + v_{100})(v_{100} - v_{95})/a}{v_{100}} = 14.4134\,\text{s}.$$

3 Demand transition and nominal region population

For intended departure times d_i, a free-flow population screen is

$$N_R(T; \tau, \delta_R) = \sum_i \mathbf{1}\{T - \tau - \delta_R < d_i \le T - \tau\}.$$

The fixed demand schedule has 4 s headway from 0 to 120 s (900 veh/h), then 8/9 s headway from 120 to 480 s (4050 veh/h). Thus the peak cohort first reaches R nominally at 194.0937 s under desired-speed insertion and 199.4356 s under zero-speed insertion. A full region-residence interval of peak arrivals begins near 208.5070 s or 213.8489 s, respectively.

Table 1: Nominal vehicles in R; these are arithmetic screens, not reference outcomes.

Decision T (s)	desired-speed insertion	zero-speed insertion
80	2	1
120	4	4
200	9	4
210	16	12
220	17	17
240	16	16
260	17	17

The old 80 s snapshot is sparse even under the faster insertion convention. The grid $T \in \{220, 240, 260\}$ s lies beyond the nominal peak-fill time under both conventions and preserves three stages for a later prefix screen. Actual counts remain audit outputs, not acceptance assumptions.

4 Why the old quota bank is poorly conditioned

At peak demand, assigning k of every six intended vehicles to the ending lane creates the following nominal lane rates; the two continuing-lane rates may swap with phase.

Ending quota $k/6$	ending lane	continuing lane A	continuing lane B
1/6	675	2025	1350
2/6	1350	1350	1350
3/6	2025	1350	675
4/6	2700	675	675

Units are vehicles per hour. SUMO's official capacity examples show that insertion throughput depends strongly on departure speed, step length, and placement; zero departure speed is specifically the weakest

example. The table above is therefore not compared to a universal capacity threshold. It does show that the unbalanced quotas impose very different lane-specific insertion burdens. Because source waiting and actual cumulative entries are part of the strong history, preparation-dependent insertion delay can reveal the preparation before the lane-drop mechanism is tested.

The cleanest small-bank structure uses one common balanced background and changes only one six-vehicle contrast block. All six permutations below give two departures to each lane and separate same-lane departures by three intended headways:

$$P_0 = (0,1,2,0,1,2), \quad P_1 = (0,2,1,0,2,1),$$
$$P_2 = (1,0,2,1,0,2), \quad P_3 = (2,0,1,2,0,1),$$
$$P_4 = (1,2,0,1,2,0), \quad P_5 = (2,1,0,2,1,0).$$

Lane 0 is the ending lane. Outside the contrast block every preparation uses P_0; from decision time onward every preparation also uses the same frozen pattern. This localizes the hidden input and holds the block's per-lane totals fixed, but it does not guarantee equal actual insertion or trajectories.

Let peak index $j = 0$ depart at 120 s with headway $8/9$ s. For the three proposed decisions, the following complete six-vehicle blocks fall inside R at decision under both nominal travel conventions:

T (s)	peak indices j	intended departure interval (s)
220	16–21	134.222–138.667
240	38–43	153.778–158.222
260	61–66	174.222–178.667

This yields $3 \times 6 = 18$ prefix preparations, down from the old $3 \times 16 = 48$, without selecting by action outcome.

5 Anticipation and exact observability

The native lane-change model prevents an analytical equality certificate. Erdmann's primary LC2013 description states that strategic lane changing may occur well before a dead lane and that urgency depends on remaining distance, current/historical speed, lane offset, and target-lane occupation. Blocked changes may alter the ego and neighboring vehicles' speeds. Those changes feed the exact per-step counts and mean-speed histories retained by the comparator.

Current official SUMO documentation lists a default `lcStrategicLookahead` of 3000 m and a left-change lookahead factor of 2. This rolling documentation is not a claim about an uninspected future execution version, but its 3000 m scale exceeds the entire 2400 m approach. Consequently, it is not defensible to assume a long upstream interval in which ending-lane assignments are dynamically irrelevant. Nor may S07 shorten lookahead or disable strategic changes merely to manufacture an exact collision.

The localized balanced block therefore improves the preparation audit without proving success. A later native prefix run must first require exact equality of the declared arrays and metadata. Any mismatch stops the action test. A failed bank would show that this contract did not produce a collision; it would not show that aggregate histories are universally sufficient.

6 Preparation-family implications

Extended road-start warm-up. Keep the accepted demand clock. Set `departPos` to `base` and `departSpeed` to `desired`; retain all insertion safety and delays, use $T = 220, 240, 260$ s, and use the

localized balanced block. This is the smallest repair and is the provisional recommendation. Its exact-match probability remains unknown.

Shift peak demand earlier. Starting the balanced peak at time zero can populate R near 90 s, reducing warm-up length. It changes the demand regime and removes the original low-demand lead-in without reducing strategic anticipation or exact-observability risk. It is therefore a broader amendment with no established scientific advantage.

Closer or nonempty initialization. Numeric mid-link departure positions are supported by the reference interface and could create a lane-blind collision immediately. In this task they introduce an artificial near-drop source or a hidden initial state, changing the known-empty/full-history contract and shifting the physical-reachability burden outside the model. This is not an admissible Candidate A repair unless DG-G1/DG-G2 are explicitly reopened.

7 Source boundary

Primary/native basis: J. Erdmann, "SUMO's Lane-Changing Model," 2015, DLR manuscript; SUMO official vehicle/type/route documentation, vehicle insertion documentation, and road-capacity methodology. Rolling documentation supports capability and design-risk statements only; no installed version or empirical scenario result is inferred.